

Verifier-Guided GRPO for Molecular Graph Reasoning in Large Language Models

Can a symbolic molecular verifier, used as a reward, teach a small open-weight model to reason over molecular structure — not just format its answers?

Model Qwen2.5-0.5B-Instruct

Reward RDKit symbolic verifier → binary correctness

Bench MolecularIQ · hard split · 849 mols / 5,111 Q

Train full fine-tune · 4xA100 · vLLM-served GRPO

BACKGROUND

Structure understanding is the bottleneck

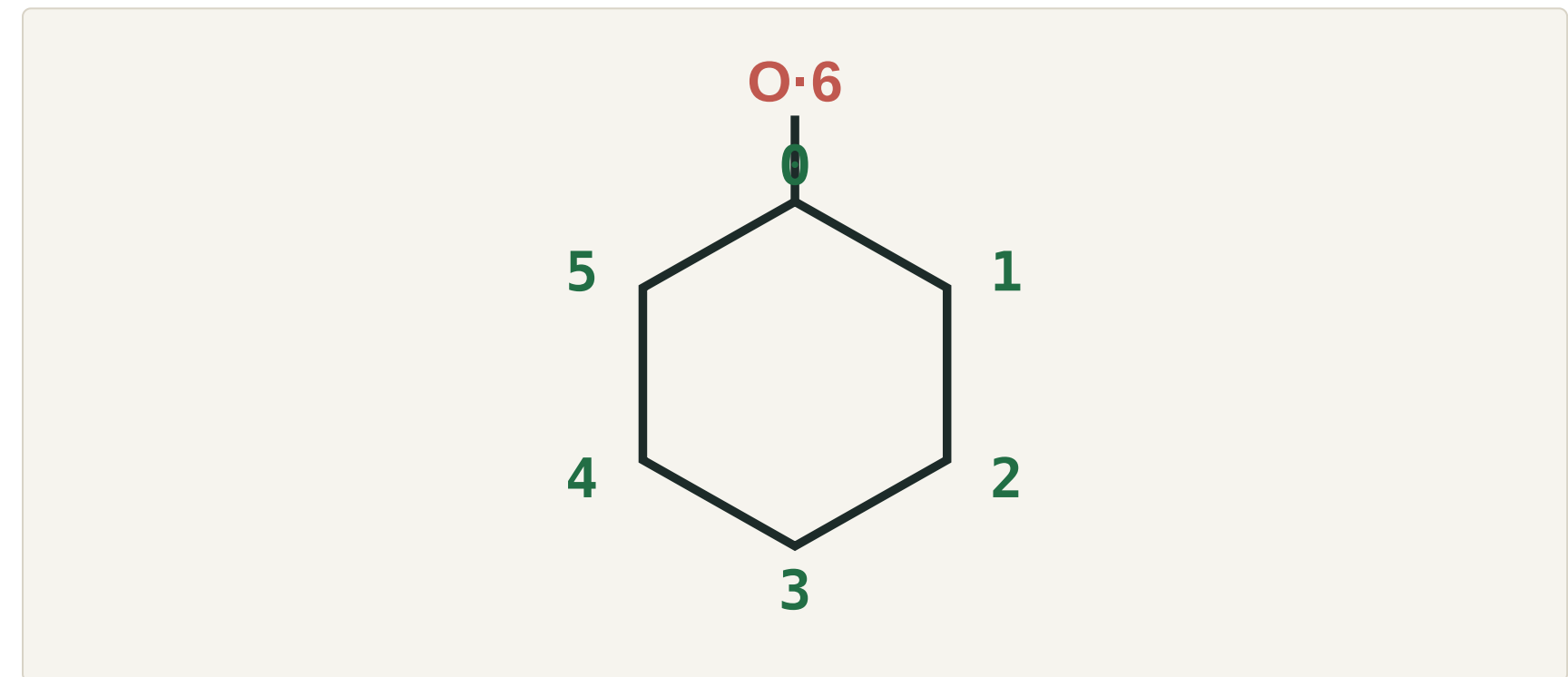
Chemistry LLMs are scored on benchmarks that leak into pretraining. **MolecularIQ** instead asks only **symbolically verifiable** questions about the molecular graph — every answer is checked exactly with RDKit, isolating reasoning from memorization.

counting

indexing

constrained generation

WHAT THE VERIFIER CHECKS

**TASK** Return the atom indices of every **oxygen** in this molecule. (0-indexed)**ANSWER** <answer>[13, 14]</answer>

✓ RDKit recomputes the ground-truth set and grants reward

1 only on an exact match — no LLM judge, no partial credit.

HYPOTHESES

H1 · MAIN

GRPO with the symbolic verifier as reward improves held-out exact-answer accuracy over prompt-only and SFT-only — most for counting & indexing.

H2 · GENERALIZATION

Gains persist on unseen molecules and randomized / kekulized / ring-relabelled SMILES, and shrink as complexity rises.

H3 · FORMAT VS REASONING

After separating extraction validity from symbolic correctness, GRPO still raises correctness — it teaches reasoning, not just <answer> compliance.

H4 · ELICITATION VS EXPANSION

Read with **pass@k**, GRPO mostly *sharpens* what the base can already reach (base catches up at large k) and *expands* only where skills compose; where exact success is unreachable it *hacks* the partial-credit reward. (see Fig 6 →)

SETUP

Four conditions

- base — minimal prompting, no system message.

HEADLINE RESULT

A verifier reward lifts a 0.5B model ~4× over prompting — SFT alone isn't enough.

On held-out MolecularIQ, **SFT+GRPO** reaches the highest exact-answer accuracy of all four conditions, with the biggest jump on counting and indexing tasks. (placeholder numbers — replace with your run)

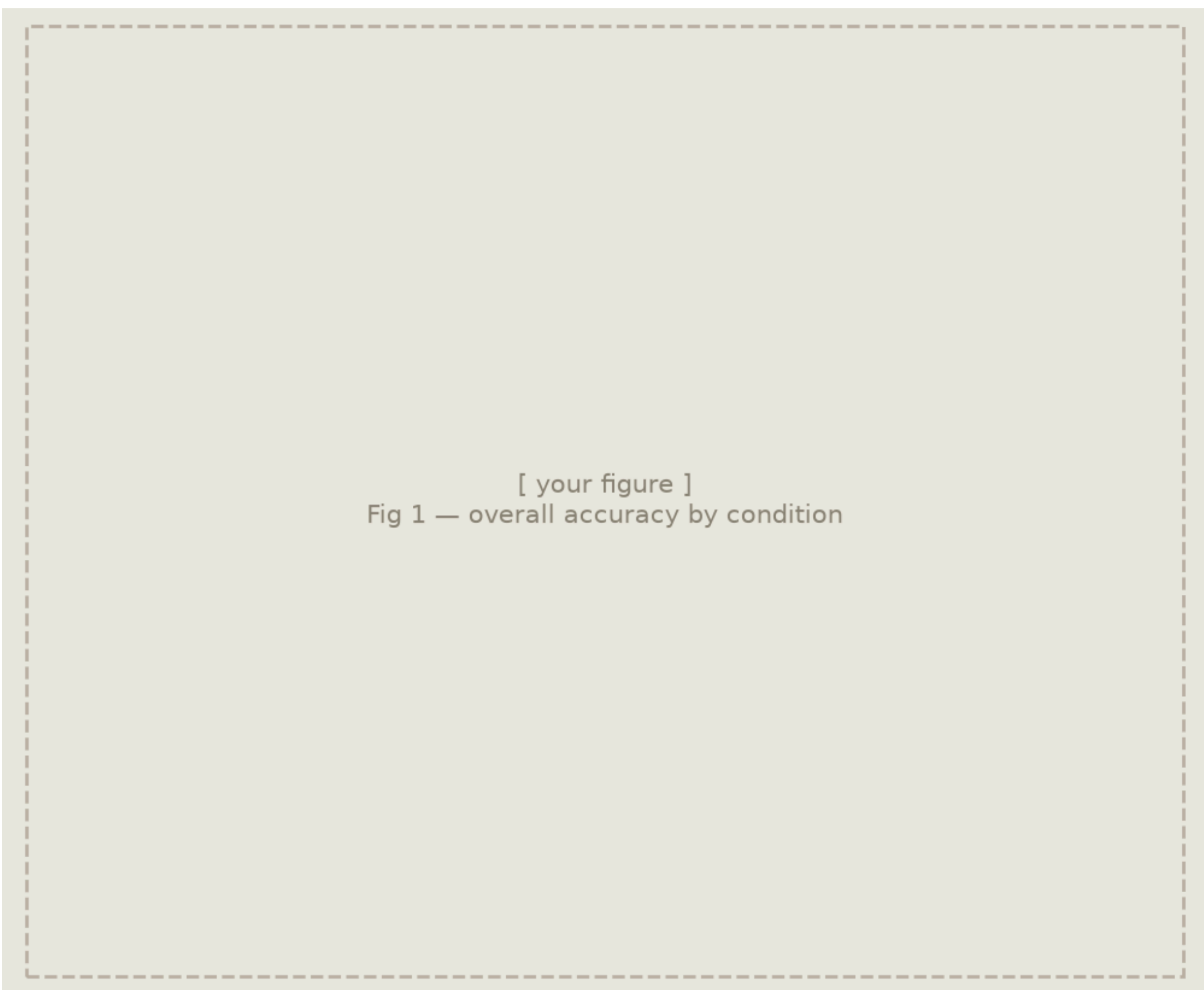


Fig 1. Overall accuracy by condition. base → prompted → SFT → SFT+GRPO on the hard test split (pass@1). Brackets mark the SFT and GRPO effects. *Synthetic placeholder — replace with your real numbers.*

WHERE THE GAINS LAND



Fig 3. GRPO helps counting & indexing most, constrained generation least (sparser reward). *Placeholder.*



Fig 4. Retention under non-canonical SMILES — GRPO keeps more of its accuracy (H2). *Placeholder.*

FORMAT VS REASONING · H3



Fig 5. The key disentangling result. Type-validity saturates early, but symbolic **correctness** keeps climbing under GRPO. The widening gap is "well-formed but wrong" — the gain is reasoning, not <answer> formatting. *Synthetic placeholder.*

ELICITATION VS EXPANSION · H4

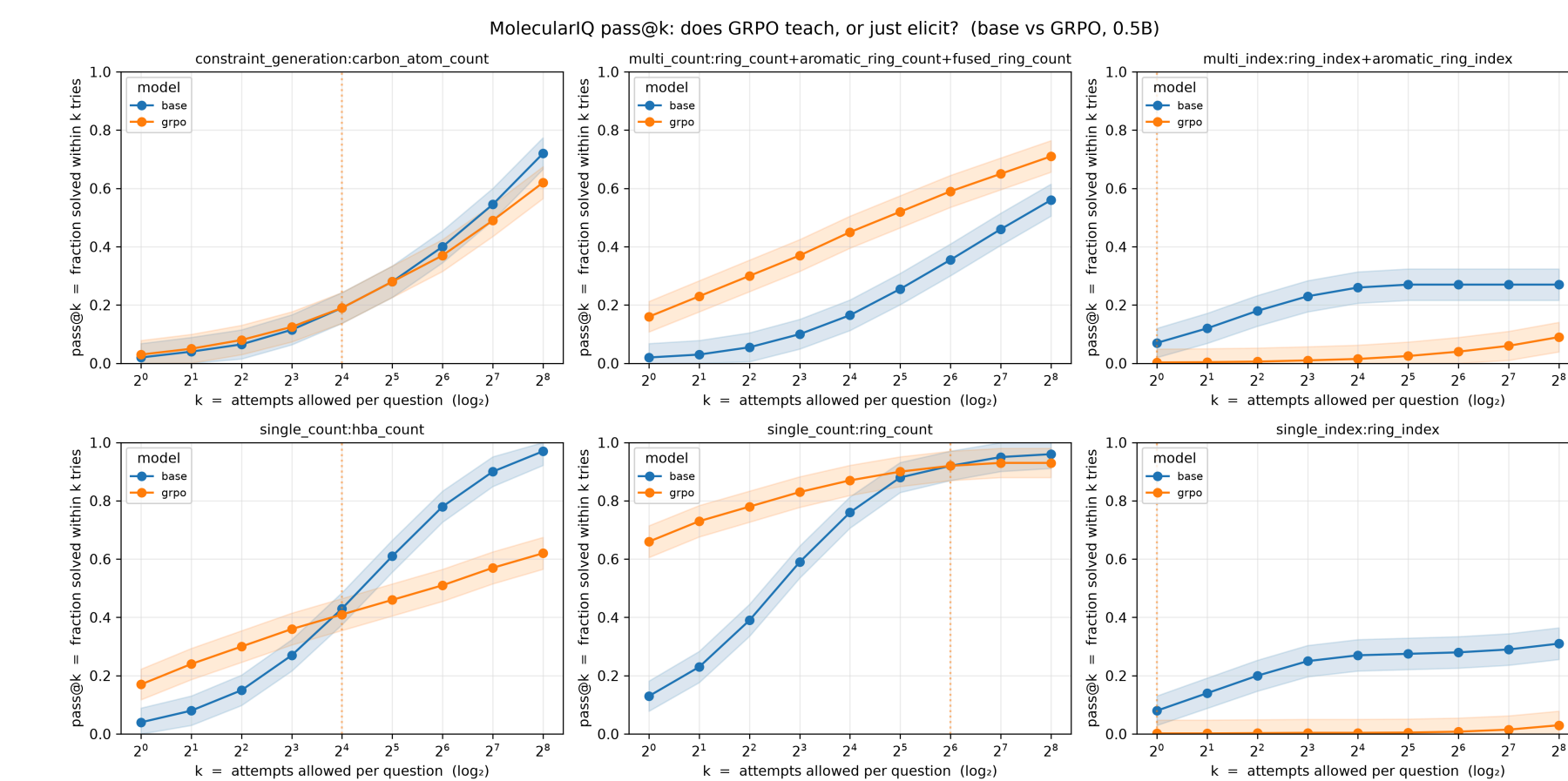


Fig 6. pass@k, base vs GRPO (0.5B). x = attempts/question (1 → 256, log_e); y = fraction solved within k tries (exact match). Base catching up at large k ⇒ *elicitation*; GRPO above base at all k ⇒ *expansion* (multi-count); GRPO below base ⇒ *reward-hacking* (indexing). *Replace with your outputs/pass_k_report figure.*

Making indexing learnable — how to fix

Index needs exact symbolic execution, so the base never samples a correct set and RL can't reinforce it → it hacks the dense proxy (wide over-covering ranges). Fixes:

- Cold-start SFT** on RDKit reasoning traces, then GRPO.
- Difficulty curriculum** — tiny molecules first, so exact reward is reachable.
- De-hack the reward** — precision/size penalty, or binary-only above a threshold.
- Tool use** — let the model call RDKit; offloads the symbolic work.

Scale helps too, but is the brute-force version of cold-start.

TAKEAWAYS

What this shows

- A symbolic verifier is a clean RL reward for molecular reasoning — no LLM judge, no leakage.
- GRPO beats SFT-only and prompting on a tiny 0.5B model under a 24 h compute budget.
- Gains survive SMILES perturbation and aren't explained by formatting — evidence of reasoning.
- pass@k (H4) shows the gain is mostly **sharpening**, with real **expansion** only where skills compose — and a reward-hacking failure where exact match is unreachable.